313335

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions :

(1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Assume suitable data, if necessary.

(6) Use of Non-programmable Electronic Pocket Calculator is permissible.

(7) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : 10

(a) Define :

(i) Latching current

(ii) Holding current

(b) Draw the symbol of IGBT and MOSFET.

(c) List the turn-ON methods of SCR.

(d) Define :

(i) Firing angle

(ii) Conduction angle

(e) Define :

(i) Duty cycle

(ii) Chopping period w.r.t. chopper

(f) State the applications of photovoltaic cell (any **2**).

(g) List any two applications of chopper.

2. Attempt any THREE of the following : 12

(a) Describe the constructional details of power MOSFET with a neat diagram.

(b) Explain the operation of R triggering with neat circuit diagram.

(c) Explain the operation of single phase fully controlled bridge rectifier with R-load.

(d) Describe light triggering method of SCR.

3. Attempt any THREE of the following : 12

(a) Explain $\frac{dv}{dt}$ protection circuit for SCR.

(b) Compare Half and Full converter (any **4** points).

(c) Explain class A commutation with suitable circuit diagram.

(d) Draw a neat circuit diagram of 1ϕ half controlled rectifier with R-L load. Write its operation.

4. Attempt any THREE of the following : 12

(a) Explain with neat circuit operation of 1ϕ full bridge inverter with R-L load.

(b) Draw the circuit diagram of pulse width modulator. Explain the working principle and draw the output waveforms.

(c) Explain operation of step-up chopper with neat circuit diagram.

(d) Describe the working of speed control of ceiling fan using TRIAC.

(e) Describe a solar power system with block diagram.

5. Attempt any TWO of the following : 12

(a) Draw the symbol and labelled V-I characteristics of following devices :

(i) SCR

(ii) TRIAC

(iii) IGBT

(b) Explain with neat circuit diagram and waveforms RC triggering method of SCR.

(c) Draw a neat circuit diagram and waveforms for half converter with R-L load and free wheeling diode. Explain its working.

6. Attempt any TWO of the following : 12

(a) Draw and explain V-I characteristics of SCR. List any two applications of SCR.

(b) Describe with neat circuit and waveforms of 1-ϕ Half bridge inverter with R-load.

(c) Describe working of step-down chopper with neat circuit diagram and waveforms. Also write formula for O/P voltage.